

From a facial scan to a better starting point.

Selecting respirators with mobile measurements
and fit-outcome data.

Edderic Ugaddan · Breathesafe LLC

Discussion with Autumn Downey

Preparing a possible contribution to the November 17 committee meeting

Make the search for fit more informed.

01

Measure

Derive facial features on a supported iPhone.



02

Prioritize

Rank mask options using facial and mask features.



03

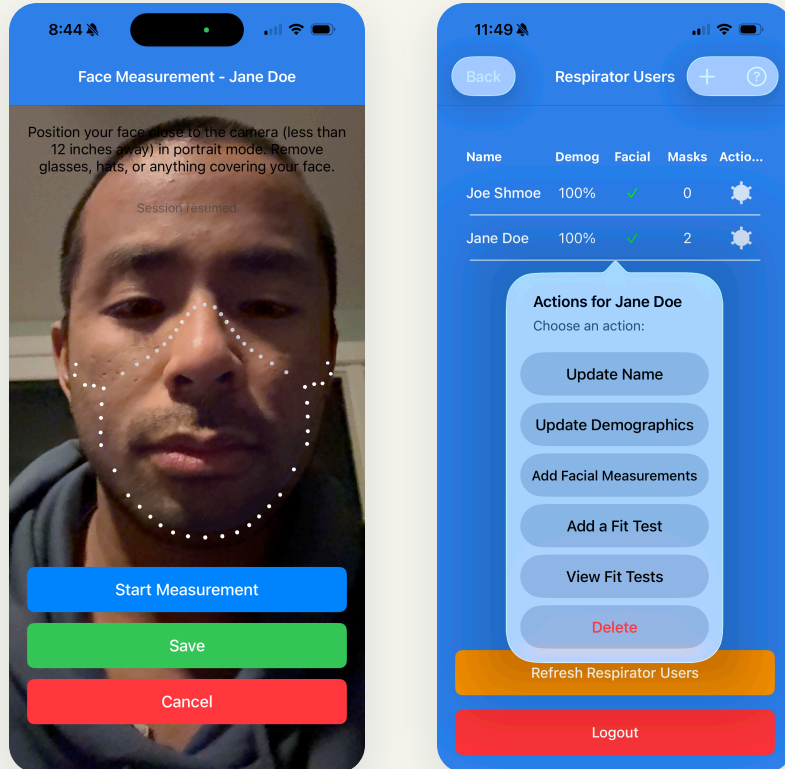
Evaluate

Try candidates and establish fit through appropriate testing.

Contributed outcomes support future model development and evaluation.

The scientific question: can this process help people find a fitting respirator with fewer unsuccessful attempts?

A working measurement and contribution workflow.



MasqFit

Captures fit-relevant facial measurements.

Breathesafe

Stores contributed outcomes and serves recommendations.

Existing demo screenshots.

Illustrate the workflow, not measurement accuracy.

A compact geometric model, learned from fit outcomes.

Measure the face

Turn selected 3D landmark pairs into distances. Average recent samples and sum nose, cheek and chin paths.



Compare each mask

Calculate facial path perimeter minus mask perimeter. Identify its fit family, style and strap type.



Score & rank

Apply coefficients learned from fit / not-fit labels, convert to a 0–1 score, and rank candidate masks.

Key assumption: perimeter mismatch captures enough relevant geometry to help prioritize options.

The current score compresses four facial path lengths into one total. It does not simulate a mask deforming or sealing against a full 3D face.

Built for the public's range of faces and choices.

Open source · GPL-3

Inspect, run, adapt and share the code. Researchers can audit methods; others can continue development.

Children included

The dataset includes children and masks sized for children, alongside adult data and options.

Broader mask choices

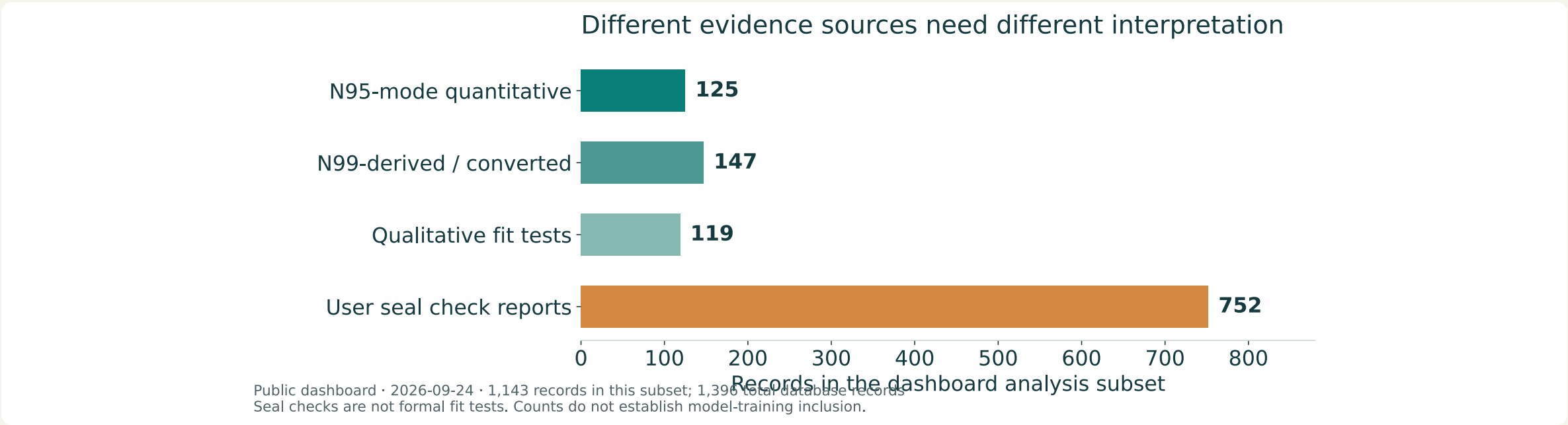
Coverage includes KN95 and FFP3 options as well as NIOSH-approved respirators.

Fit + breathability

Connect fit evidence with breathing-resistance data to help people find options they can comfortably wear.

A broader community-use scope: families, everyday use and international mask options, alongside occupational applications.

The dataset contains several kinds of evidence.



55

users in demographic aggregation

757

analyzed user–mask pairs

83 / 310

catalog entries with fit-related results

A mask needs to fit—and be practical to keep wearing.

Lower breathing resistance

At the same airflow and test setup, a lower pressure drop means air passes through more easily.

Breathability supports comfort and can make sustained wear easier. Fit, filtration and comfort all matter.

Compare measurements at matched test conditions; pressure drop alone does not determine comfort.

DATA CONTRIBUTION

Thank you, Gerard Hughes.

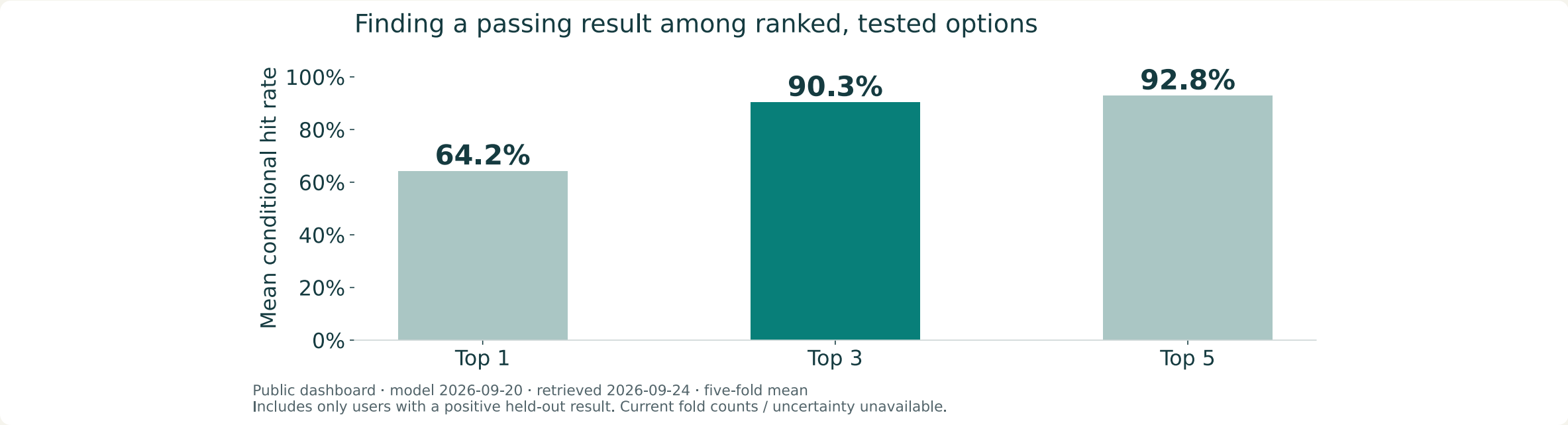
For contributing breathability measurements to the dataset.

156 / 310

Catalog entries with breathability data in the September 24 snapshot.

The project records breathability alongside fit-related evidence, making wearability part of the information available to users.

Promising ranking evidence, with a defined scope.



Includes only users with a positive held-out result; ranks previously tested observations.
Success after sending new users three masks remains a prospective research question.

Three separate questions about validity.

1

Measurement

How accurate and repeatable are phone-derived dimensions?

Benchmark against a reference method across devices and face shapes.

2

Prediction

How well do rankings generalize to new people and mask options?

Separate formal test outcomes from weaker labels; assess calibration.

3

Practical benefit

Does using the app improve selection in a real workflow?

Prospectively compare outcomes, time and unsuccessful attempts.

Internal retrospective evaluation addresses only part of this chain.

Design choices shape who can benefit.

Access and coverage

Current scanning workflow uses supported iOS hardware.

Device access, sparse mask coverage and missing demographic data constrain generalization.

29 of 55 users in the dashboard age aggregation have unspecified / undisclosed ages.

Data handling

Baseline policy: derived numerical measurements; no retained face images, video or full 3D scans.

Opt-in research can retain a limited subset of facial measurement points.

Numerical measurements still deserve careful handling.

Test the decision people actually need to make.

Enroll

New participants; a predefined respirator catalog and protocol.



Compare

App-guided selection versus a predefined usual-selection approach.



Measure

Formal fit-test outcomes for candidate masks.

Primary endpoint

Fraction of participants finding at least one passing option within a fixed candidate budget.

Supporting endpoints

Attempts, time, scan failures, calibration and performance across predefined groups.

A protocol-development proposal: sample size, allocation and test procedures require study design.

Facial scanning for respirator selection: what works, what needs validation.

Demonstrate

A concrete scan-to-recommendation workflow and its open implementation.

Explain

How outcome data become rankings, and what existing evaluation does and does not establish.

Invite scrutiny

Measurement benchmarks, credible study endpoints and data gaps.

Connection to custom-fit facepieces: shared measurement and validation questions. Current project work centers on selecting existing mask options.

What would be most useful to the committee?

1. Which decisions or scientific questions should the November session inform?
2. How much should I emphasize the scanning method, the recommender, or validation?
3. Who else will speak, especially on custom-fit fabrication?
4. What talk length, demonstration format and advance materials would help?

Desired next step: agree on a focused contribution and the evidence it should include.

Several evidence paths feed a fit / not-fit label.

Evidence source	Project labeling rule
Direct N95-mode quantitative test	Aggregate exercise fit factors; label pass at ≥ 100 .
N99-mode quantitative test	Estimate leakage using filtration data; aggregate and apply the project's $\times 2$ adjustment; label pass at ≥ 100 .
Qualitative fit test	Use recorded exercise outcomes, subject to the service's missing-result handling.
User seal check	Failure $\rightarrow$ negative label. Passing or incomplete check $\rightarrow$ no label from this source.

“Fit” here is a data label. Preserve whether it was measured, inferred, or reported.

When sources disagree for the same record ID, current consolidation prioritizes a negative label. Unlabeled records are excluded by default.

The same ratio, different particles counted.

Fit factor = outside particle concentration / inside particle concentration

N99 mode / standard mode

Counts a broad aerosol size range. The in-mask count can reflect both filter penetration and leakage.

With sufficiently efficient filters, penetration contributes less and the ratio is more specific to fit.

N95 mode / N95-Companion

Selects a narrow band of negatively charged particles, nominally around 55 nm, that suitable respirator filters capture efficiently.

Reducing the filter-penetration contribution makes the measurement more specific to leakage.

Mode names describe the measurement method; they do not certify a mask as N95 or N99.

Select particles in the instrument's sample stream.

Sample

Use a mask probe and an ambient inlet. Switch between samples, allowing the lines to purge.



Select & count

The classifier selects particles by electrical mobility. The condensation particle counter measures their concentration.



Compare

Compare outside and inside concentrations during the prescribed fit-test exercises.

Why this helps N95-class filters

It reduces false fit failures caused by particles penetrating acceptable filter media.

Keep the interpretation specific

The mode estimates fit; it is not a filter-certification test or an all-particle protection factor.

Room air is not prefiltered before the wearer breathes it. Use the supported respirator category, instrument checks and prescribed protocol.

Estimate what gets through the filter itself.

Aaron Collins (“MaskNerd”)

Credit to **Aaron Collins of OpenAeros** for contributed filtration-efficiency data used by the project.

Imported entries identified as Aaron’s data are one source. Source measurements are not proof of fit for another wearer.

[Aaron Collins · OpenAeros](#)

Contributed sealed measurements

An optional N99-mode “Normal breathing (SEALED)” observation estimates penetration with face-seal gaps minimized.

Residual leakage, breathing flow, aerosol conditions and mask variation can affect this estimate.

Sealed ratio $S = 1 / (1 - E)$ · E = filtration efficiency as a fraction

Example: $E = 98\% \rightarrow S = 50$. The code pools available sealed ratios by mask using an arithmetic mean.

Separate filter penetration from inferred leakage.

A simplified mixture

Let F be the observed N99-mode ratio, E the estimated filtration efficiency, and L the fraction entering through leaks.

$$1/F = L + (1 - L)(1 - E)$$

Assumes leakage admits ambient particles and the filtered stream has penetration $1 - E$, under comparable sampling conditions.

Solve for leakage

$$L = 1 - (1 - 1/F) / E$$

$$\text{Estimated leakage fit factor} = 1/L$$

This is the core mathematical estimate. The production labeling formula adds project-specific adjustments.

Illustrative: $E = 98\%$, $F = 25 \rightarrow L \approx 2.04\% \rightarrow \text{estimated leakage fit factor} \approx 49$.

As F approaches the sealed reference, the estimate becomes highly sensitive to measurement error.

[Equation derivation: inferring-n95-mode-fit-factors.ipynb](#) ↗

From the leakage estimate to the stored label.

1. **Choose a sealed reference.** Average available sealed ratios S . If the largest exercise ratio is $\geq S$, replace S with that maximum + 1.
2. **Convert each eligible exercise.** Set $E = 1 - 1/S$; compute $1 / [1 - (1 - 1/F) / E]$. Exclude “Talking” and the sealed reference exercise.
3. **Aggregate and adjust.** Take the harmonic mean of the converted factors, multiply by **2**, then label pass at **≥ 100** .

$$\text{Stored estimate} = 2 \times n / \Sigma(1 / \text{converted exercise factor})$$

If every exercise in the illustrative example gives 49, the stored estimate is 98 → fail.

The notebook calls $\times 2$ a sensitivity-oriented bias adjustment. It is not part of the physical derivation or the direct N95 measurement.

Failure supplies a negative signal; a pass is not proof.

Failed check

“Too big,” “Too small,” or a completed response reporting “A lot of air movement.”

→ **not-fit label**

An inferred negative outcome, not a newly performed formal fit test.

Passed check

No reported failure under the questionnaire’s criteria.

→ **no positive fit label**

The check cannot establish that a formal fit test would pass.

Incomplete check

Missing or uninterpretable required answers.

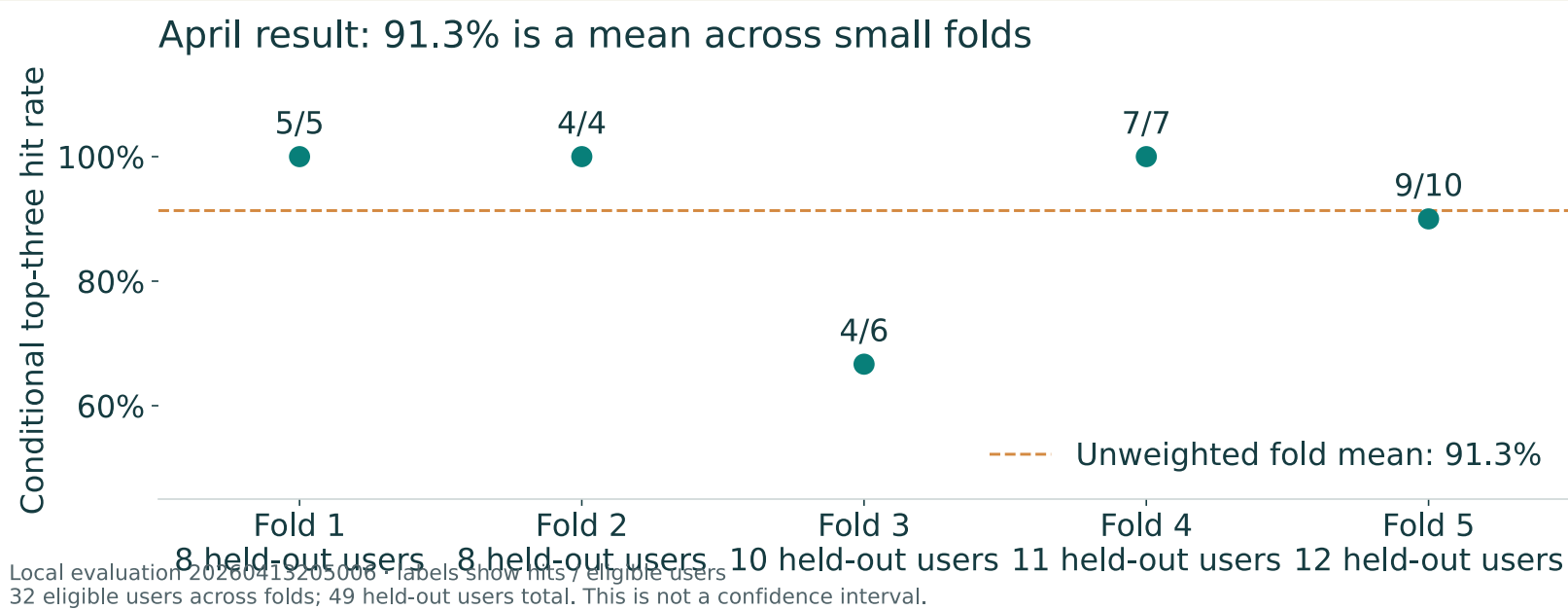
→ **unknown**

No label from this source; another recorded test may provide one.

In this dataset: user seal check fail → failed-fit label.

User seal check pass $\nRightarrow$ successful fit test.

The earlier “91%” needs its denominator.



A matching April run also reports holdout AUC 0.786: 337 rows from 10 users.
The old “n=8” label does not describe the complete cross-validation population.

The freedom to inspect, improve and carry it forward.

Use & inspect

Run the software, study its source, and evaluate how the methods work. Commercial use is permitted.

Modify & share

Adapt or fork the code. Distributed covered derivatives must preserve GPL-3 freedoms and license terms.

Provide source

When distributing covered binaries, provide recipients access to the corresponding source under the license's rules.

Practical advantages: auditable methods, independent experimentation, and continuity if the original maintainer steps away.

GPL-3 covers the software; it does not automatically make participant data public. Private changes and merely running a hosted service do not generally trigger a public-source release obligation.

Different scopes, with a distinct public-facing contribution.

Project	Scope established here	How MasqFit contributes
FitMask · Mobomo	Available for iPhone; listing advertises facial-feature matching to N95 and KN95 sizes.	MasqFit combines open code, contributed fit outcomes, child-inclusive data and breathability information.
Planned NIOSH apps	Android + iPhone; adult / N95 focus, as described in project-owner and NPPTL discussions.	MasqFit broadens coverage to children, child-sized masks, and options such as KN95 and FFP3.
MasqFit + Breathesafe	Current iOS scanning workflow; GPL-3 source; broader community-use dataset and mask coverage.	Designed to help the general public navigate both fit and practical wearability.

This is a scope comparison, not a head-to-head accuracy study. FitMask already includes KN95. Unverified competitor features are not treated as absent.

Sources: [FitMask App Store listing](#); April NPPTL meeting notes and project-owner confirmation. NIOSH release dates and final feature set not verified.

Measure selected segments of the facial mesh.

1. Landmark coordinates

ARKit provides indexed 3D vertices. The app selects predefined pairs along fit-relevant facial paths.

Point $i = (x_i, y_i, z_i)$

2. Euclidean segment lengths

$$d_{ij} = \sqrt{[(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2]}$$

Compute each pair's length per sample, then average the recent lengths. Convert meters to millimeters.

The current measurement engine averages the last 10 samples. The controller normally collects 30 updates before auto-stopping.

Assumptions: stable landmark correspondence, meaningful scale and a steady scan. Averaging can reduce fluctuation; it cannot remove systematic measurement bias.

Five aggregates are collected; four enter the score.

Aggregate	Construction	Current fit score
Nose	8 predefined segment lengths	Included in perimeter total
Upper cheek	14 segment lengths	Included in perimeter total
Mid-cheek	10 segment lengths	Included in perimeter total
Chin	14 segment lengths	Included in perimeter total
Strap-related facial path	8 segment lengths	Collected; not used by current score

$P_{\text{face}} = \text{nose} + \text{upper cheek} + \text{mid-cheek} + \text{chin}$

These are sums of short 3D chords along chosen paths, not a full head circumference or a contact-pressure measurement.

Sources: FacialMeasurement.aggregated_arkit_measurements; FacialAggregates; train.py FACIAL_MEASUREMENTS and prep_data_in_torch_with_categories.

A quadratic mismatch model with a logistic output.

$$\Delta = (P_{\text{face, mm}} - P_{\text{mask, mm}}) / 10 \quad \cdot \quad \text{score} = 1 / (1 + \exp(-z))$$

$$z = (a_{\text{family}} + a_{\text{style}})\Delta^2 + (b_{\text{family}} + b_{\text{style}})\Delta \\ + c_{\text{family}} + c_{\text{style}} + s_{\text{strap type}}$$

Family: related variants

A project-assigned fit group. Example: Trident Small in white and black sharing one family (illustrative grouping).

Style: overall shape

Examples: **Boat** (trifold), **Bifold**, **Cup** and **Duckbill**.

Strap type: attachment

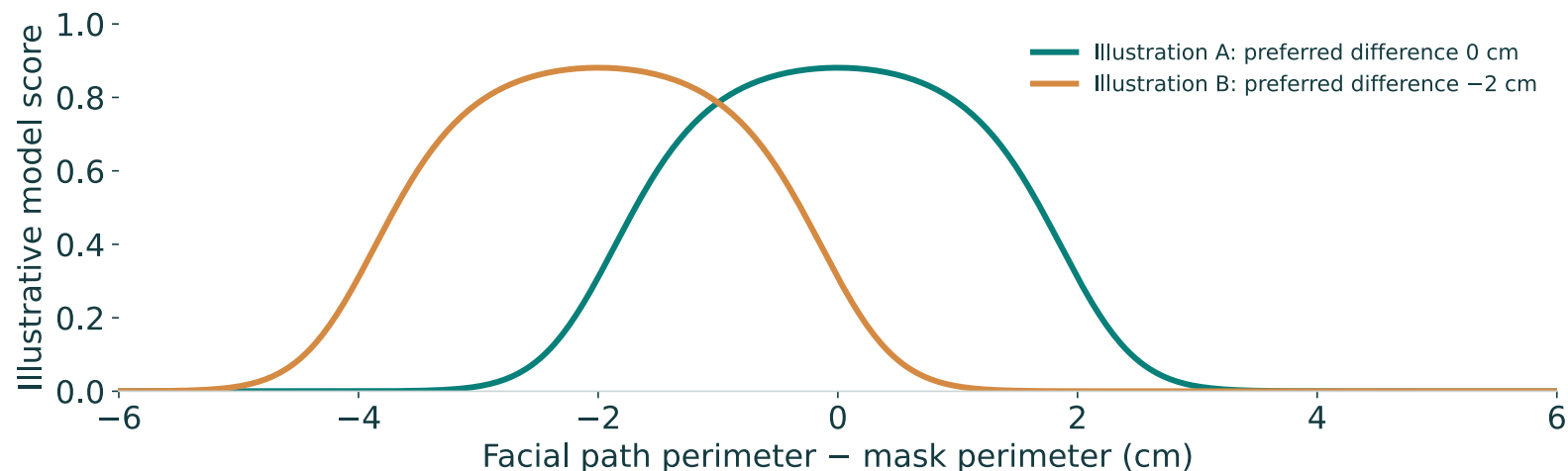
Headstrap or **Earloop**, with separate **Adjustable** categories. This is not the facial strap-path length.

Family and style each supply curvature, slope and intercept; strap type adds an offset.

New training constrains curvature with $a = -\exp(\text{raw parameter})$.

Each family / style favors a range of mismatch.

The model allows a preferred mismatch, not only an exact match



Schematic only · invented coefficients: $\text{score} = \text{sigmoid}[2 - 0.7 \times (\text{difference} - \text{preferred difference})^2]$
Not actual mask predictions, fitted coefficients, participant data or validation results.

A mask can prefer a nonzero difference: the learned linear term shifts the peak.

Negative curvature penalizes moving far enough to either side of that preferred difference.

For a fixed mask and strap type, two faces with the same summed perimeter receive the same score.

Fit the coefficients, then score a new face.

Build training rows

Join measurements, mask metadata and fit / not-fit labels. Some missing scan aggregates are estimated from traditional measurements.



Learn parameters

Optimize binary cross-entropy on labeled rows using Adam. Evaluate with user-grouped splits.



Serve recommendations

Use saved parameters and mask metadata to compute scores for candidate masks; return them in descending score order.

The model learns associations with the constructed outcome labels. It inherits their measurement error, selection effects and source differences.

Grouping the fit-model split by user is helpful, but end-to-end evaluation must also assess imputation and preprocessing within the split.

What this compact model captures—and leaves out.

Geometry compression

One perimeter proxy carries the facial signal. Different nose/chin proportions with the same total are indistinguishable to this score.

Shared effects

Fit-family and style effects add together; strap type shifts the score. Actual strap tension, deformation and donning technique are not simulated.

Data transfer

Labels, imputed measurements and contributed users must generalize to new people. Repeated observations can give some users more influence.

Score interpretation

A sigmoid supplies a 0–1 score, not automatic calibration or uncertainty. Breathability and demographics are not direct inputs to this scoring equation.

Next validation targets: scan repeatability, perimeter compression, imputation, unknown-mask behavior and prospective fit outcomes.

Keep the claims connected to their evidence.

Project evidence

[April 14 CDC/NIOSH/NPPTL presentation](#)

[September 24 public dashboard snapshot](#)

Model timestamp: September 20, 2026

[Breathesafe source](#) · [MasqFit source](#)

[Project privacy policy](#)

Meeting and scientific context

[NASEM standing committee](#)

[CDC: fit testing](#)

Interpretation

Retrospective ranking results do not establish scan accuracy, calibrated individual probabilities, or prospective fit success.